

2.28 For a formula ϕ built up using the connectives $\neg, \wedge, \vee$, let ϕ^* be constructed by replacing each propositional variable in ϕ by its negation. So if ϕ is the formula $((q \vee p) \wedge \neg p)$, ϕ^* is $((\neg q \vee \neg p) \wedge \neg \neg p)$.

- (a) For any truth assignment v , let v^* be the truth assignment which gives each propositional variable the opposite value to that given by v , i.e.

$$v^*(p) = \begin{cases} T, & \text{if } v(p) = F, \\ F, & \text{if } v(p) = T, \end{cases}$$

for all propositional variables p . Show that $v(\phi) = v^*(\phi^*)$. [*Hint*: This is really a statement about all formulas ϕ of a certain sort, so what is the likely method of proof?]

- (b) (i) Use the result of part (a) to show that ϕ is a tautology if and only if ϕ^* is a tautology.
(ii) Is it true that ϕ is a contradiction if and only if ϕ^* is a contradiction? Explain your answer.

proof of (a). We proceed by induction on the length of a formula

Base Case: Suppose we have a formula, ϕ , of length zero. So then ϕ is just a propositional variable, p . WLOG, let $v(p) = T$. Then $\phi^* = \neg p$. So then $v^*(\phi^*) = v^*(\neg p) = T$.

Induction Hypothesis: For all formulas, ϕ , of length $\leq n$, $v(\phi) = v^*(\phi^*)$.

Inductive Step: Let ϕ be a formula of length $n + 1$. Since $n + 1 \geq 0$, we know that ϕ contains connective, so it isn't a propositional variable, so by the definition of formula it is one of these forms: $\neg\theta, (\theta \wedge \psi), (\theta \vee \psi)$.

Case $\phi = \neg\theta$:

Then θ has length $\leq n$, so by the inductive hypothesis $v(\theta) = v^*(\theta^*)$.

By the definition of $*$, $\phi^* = \neg(\theta^*)$, so we want to show $v(\neg\theta) = v^*(\neg(\theta^*))$.

Since the truth table for $\neg$ is symmetric under interchanging T and F , WLOG suppose that $v(\theta) = v^*(\theta^*) = F$.

So then $v(\phi) = v(\neg\theta) = T$.

Since $v^*(\theta^*) = F$, $v^*(\neg(\theta^*)) = T = v^*(\phi^*)$.

And so, $v(\neg\theta) = v^*(\neg(\theta^*))$, i.e., $v(\phi) = v^*(\phi^*)$.

Case $\phi = (\theta \vee \psi)$:

Then θ and ψ have length $\leq n$, so the the inductive hypothesis $v(\theta) = v^*(\theta^*)$ and $v(\psi) = v^*(\psi^*)$.

By the definition of $*$, $\phi^* = (\theta^* \vee \psi^*)$, so we want to show $v((\theta \vee \psi)) = v^*((\theta^* \vee \psi^*))$.

Since the truth table for $\vee$ is symmetric under interchanging T and F , WLOG suppose that $v(\theta) = v^*(\theta^*) = T$ and $v(\psi) = v^*(\psi^*) = F$.

So then $v(\phi) = v((\theta \vee \psi)) = T$.

But (this is where i get confused) $v^*(\phi^*) = v^*((\theta^* \vee \psi^*)) = F$

□